Jia et al. 2024 — MetaboAgeMort

One-Page Summary

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Core Specifications

Attribute	Value
Clock Name	MetaboAgeMort
Sample Size	239,291 individuals (UK Biobank)
Platform	NMR spectroscopy (Nightingale Health)
Input Biomarkers	35 metabolic biomarkers + chronological age (selected from 249 via LASSO Cox)
Training Target	10-year all-cause mortality (Gompertz proportional hazards)
Age Range	Median 58.3 years (IQR: 50.6–63.7)
Train/Test Split	70% training (n=167,506) / 30% testing (n=71,785)

Performance

Metric	Value
Age Correlation	r = 0.85 (overall); r = 0.85 (female), r = 0.86 (male)
All-Cause Mortality HR	HR = 1.07 per year increase (95% CI: 1.06–1.08)
C-statistic Improvement	+0.017 over conventional risk factors ($p < 0.001$)
IDI	IDI = 0.018 (95% CI: 0.004–0.021) beyond conventional model
Disease Associations	11 diseases: cancer, CVD, dementia, CKD, T2DM, liver, respiratory, hypertension, depression, anxiety, eye disease

Unique Features

- Mortality-trained clock—trained on 10-year all-cause mortality, not chronological age, better capturing biological aging heterogeneity
- MetaboAgeMortAccel delta — residual of MetaboAgeMort regressed on chronological age; quantifies biological age acceleration/deceleration
- Cause-specific mortality—significant associations with cancer, CVD, respiratory, digestive, and neurodegenerative disease mortality
- 99 modifiable factors — systematically identified across 7 categories (pulmonary function, body composition, SES, diet, smoking, alcohol, disease status)
- GWAS — 99 genomic risk loci and 271 genes identified; liver tissue enrichment highlighted
- Outperforms prior clocks—higher AUC than MetaboAge and MetaboHealth for 10-year all-cause mortality ($p < 0.001$)

Top Contributing Biomarkers (selected via LASSO Cox)

#	Biomarker	Category
1	Glycoprotein acetyls (GlycA)	Inflammation
2	Albumin	Fluid balance
3	Amino acids (Ala, His, Leu, Val, Phe, Tyr)	Amino acids
4	Ketone bodies (3-HB, acetate, acetoacetate, acetone)	Energy metabolism
5	VLDL, HDL, LDL subclass ratios	Lipoproteins (extensive)
6	Linoleic acid, omega-3 FA, MUFA ratios	Fatty acids
7	Glucose, pyruvate, citrate	Glycolysis
8	Creatinine	Fluid balance / renal

Strengths vs. Limitations

Strengths	Limitations
• Largest mortality-targeted metabolomic aging study • Comprehensive disease associations (11 conditions) • 99 modifiable factors identified for intervention • GWAS included (99 loci, 271 genes) • Outperforms MetaboAge and MetaboHealth • Concentration-unit biomarkers (cross-cohort applicable)	• No external validation cohort (UK Biobank only) • Nightingale NMR platform required • Coefficients not directly published (need to request) • Age range 40–70 years; UK/European population • EDTA plasma required (serum not validated) • Poor for neurological disease prediction

Comparison to Mutz 2024 (MileAge) and Zhang 2024

Feature	Jia 2024 (MetaboAgeMort)	Zhang 2024	Mutz 2024 (MileAge)
Goal	Mortality prediction	Mortality prediction	Age prediction
Biomarkers	35 (+CA)	54 ⭐	168
Sample Size	239,291 ⭐	250,341 ⭐	225,212
Age Corr.	r = 0.85	r = 0.29	r = 0.87 ⭐
Mortality HR	HR = 1.07/year	AUC = 86.8% ⭐	HR = 1.51/SD
GWAS	Yes ⭐ (99 loci)	Yes ⭐	No

Modifiable Factors	Yes ⭐ (99 factors)	Limited	No
Algorithm	Gompertz + LASSO Cox	LASSO Cox	Cubist (17 tested) ⭐

When to Use

- Use for: 10-year mortality prediction | Identifying modifiable aging factors | Intervention study design | GWAS/genetic analysis of metabolic aging
- Consider for: Comparing MetaboAgeMort acceleration across lifestyle/environmental subgroups
- Don't use for: Direct age estimation (use MileAge) | Short-term 1-year mortality (use Zhang 2024) | Non-Nightingale NMR platforms | Non-EDTA plasma

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RECOMMENDED FOR: Mortality prediction with genetic/modifiable factor analysis

Why: Unique among metabolomic clocks for systematically identifying 99 modifiable aging factors and GWAS loci. Best choice when the research question extends beyond prediction to understanding aging drivers and designing interventions.